Chapter 7: Bandits and Policy-Gradient Reinforcement Learning

Part III — Statistical Foundations of Deep Learning & Reinforcement Learning

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Abstract

Two statistical problems sit under reinforcement learning's action side. The first is the explore–exploit trade-off, in its purest form the **multi-armed bandit**: pull arms to maximise reward while learning which arm is best, and the right amount of exploration falls out of a concentration inequality (UCB) or a posterior (Thompson sampling), with regret that grows only logarithmically. The second is *optimising* a policy you can only sample from — and its answer is the **policy gradient**, which is nothing but the score function of Chapter 2 applied to a policy. The gradient of expected reward equals the expected reward times the score, an identity that lets you differentiate through a sampling step, and subtracting a baseline — a control variate — cuts its variance without biasing it.

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1 Bandits: exploration from a confidence bound

A **K-armed bandit** offers arms with unknown mean rewards $\mu_1, \dots, \mu_K$; each round you pull one and see a noisy reward. The cost of not knowing the best arm $\mu^\star = \max_k \mu_k$ is the **regret** $\mathrm{Reg}(T) = \sum_{t=1}^{T} (\mu^\star - \mu_{a_t})$. Pulling greedily can lock onto a wrong arm forever (linear regret); exploring at a constant rate (ϵ-greedy) wastes a constant fraction of pulls, also linear. The art is to explore *just enough*, and Hoeffding says how much.

Definition 1.1 (Upper confidence bound)**.** After n_k pulls of arm k with empirical mean $\hat{\mu}_k$, the **UCB** index is

$$\hat{\mu}_k + \sqrt{\frac{c \log t}{n_k}},$$

and UCB1 pulls the arm of largest index.

The bonus is exactly a Hoeffding confidence width (Chapter 1): with high probability $\mu_k \le \hat{\mu}_k + \sqrt{c \log t / n_k}$, so the index is an *optimistic* estimate of the arm's value. Optimism forces exploration of under-sampled arms (small n_k, large bonus) but lets the bonus shrink as evidence accumulates, which yields **logarithmic** regret, $\mathrm{Reg}(T) = O\big(\sum_{k:\Delta_k>0} \frac{\log T}{\Delta_k}\big)$ with $\Delta_k = \mu^\star - \mu_k$. **Thompson sampling** reaches the same logarithmic regret by a Bayesian route: keep a posterior over each arm's mean, draw one sample from each, and pull the arm with the largest sample — arms the posterior is unsure about are sampled optimistically on their own. The left panel of the figure shows both staying sub-linear while ϵ-greedy's regret grows as a straight line and eventually overtakes them.

2 The policy gradient is the score function

Now the harder problem: adjust a parameterised policy $\pi_\theta(a \mid s)$ to maximise expected reward $J(\theta) = \mathbb{E}_{a\sim\pi_\theta}[R(a)]$. The reward depends on θ only through the *sampling* distribution, so we cannot differentiate R — but we can differentiate the probability, with the log-derivative trick of Chapter 2.

Theorem 2.1 (Policy gradient / REINFORCE)**.** $\nabla_\theta J(\theta) = \mathbb{E}_{a\sim\pi_\theta}\big[R(a)\,\nabla_\theta \log \pi_\theta(a)\big]$.

Proof. $\nabla_\theta J = \nabla_\theta \sum_a \pi_\theta(a) R(a) = \sum_a R(a)\,\nabla_\theta \pi_\theta(a)$. Multiply and divide by $\pi_\theta(a)$ and use $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ (the log-derivative identity):

$$\nabla_\theta J = \sum_a \pi_\theta(a)\, R(a)\, \nabla_\theta \log \pi_\theta(a) = \mathbb{E}_{a\sim\pi_\theta}\big[R(a)\,\nabla_\theta \log \pi_\theta(a)\big].$$

□

The quantity $\nabla_\theta \log \pi_\theta(a)$ is the **score** — the exact object from Chapter 2's maximum likelihood, now of a policy. The expectation is estimated by sampling: take an action, observe its reward, and nudge θ in the direction $R(a)\nabla \log \pi_\theta(a)$. This is **REINFORCE**, and it turns "optimise through a sampler" into an ordinary stochastic gradient ascent (Chapter 3).

3 Baselines are control variates

REINFORCE is unbiased but noisy, because $R(a)$ can be large and its scale inflates every gradient. The fix is a **baseline** b: subtract it from the reward.

Theorem 3.1 (A baseline does not bias the gradient)**.** For any constant b, $\mathbb{E}_{a\sim\pi_\theta}\big[(R(a) - b)\nabla \log \pi_\theta(a)\big] = \nabla_\theta J(\theta)$.

Proof. It suffices that the subtracted term vanishes: $\mathbb{E}[b\,\nabla \log \pi_\theta(a)] = b \sum_a \pi_\theta(a) \nabla \log \pi_\theta(a) = b \sum_a \nabla \pi_\theta(a) = b\,\nabla \sum_a \pi_\theta(a) = b\,\nabla 1 = 0$, using the mean-zero-score identity of Chapter 2. □

So the baseline is a **control variate**: it changes the variance of the estimator but not its mean, and the variance-minimising b is a reward-magnitude weighted average of the returns (close to, but not exactly, the mean reward). The middle panel shows REINFORCE with a baseline learning faster and more smoothly than without; the right panel plots the estimator's variance against b, a clean parabola minimised near the mean reward. In full RL the natural baseline is the *value function* $V(s)$ (Chapter 6), and $R - V(s)$ is the **advantage** — the actor–critic architecture is exactly this pairing. Finally, when the data come from a different behaviour policy μ, the same log-derivative machinery corrects for it by **importance sampling** (Chapter 4), reweighting each sampled return by $\pi_\theta(a)/\mu(a)$ — **off-policy** policy gradients.

Summary

In a K-armed bandit, regret is the price of not knowing the best arm; greedy and ϵ-greedy give linear regret, while UCB (an optimistic Hoeffding confidence bound, Chapter 1) and Thompson sampling (a posterior draw) give logarithmic regret. To optimise a sampled policy, the policy-gradient theorem gives $\nabla_\theta J = \mathbb{E}[R(a)\nabla \log \pi_\theta(a)]$ via the log-derivative trick, where $\nabla \log \pi_\theta$ is the Chapter 2 score; REINFORCE is the stochastic-gradient estimate. A baseline b leaves the gradient unbiased (mean-zero score) but cuts its variance — a control variate — with the value function as the natural choice ($R - V$ is the advantage, i.e. actor–critic), and importance sampling (Chapter 4) extends it off-policy.